

Applied Statistical Analysis in R

Anxiety intervention, film preferences, weight-loss programmes, and heart-health outcomes

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Prepared from R Markdown files and R code

Statistical methods: independent-samples t-test, chi-square analysis, one-way ANOVA, and factorial ANOVA

Analysis overview

Four applied studies using psychological and health-related datasets

This report presents four analyses conducted in R. Each section defines the research question, summarises the data, reports the statistical result, interprets the visual evidence, and states the practical conclusion. Statistical significance is evaluated at $\alpha = .05$, while effect sizes are used to distinguish statistical detection from practical importance.

Studies included

Study	Design	Primary method	Outcome
Air-traffic-controller anxiety	Two independent intervention groups	Independent-samples t-test	Post-intervention anxiety score
Personality and film preference	Two categorical variables	Chi-square test of independence	Preferred film genre
Diet and weight loss	Four independent programmes	One-way ANOVA	Weight loss after 8 weeks
Fasting, exercise, and heart health	4 x 2 factorial design	Factorial ANOVA	Change in heart-health score

Interpretive approach

- Report exact group means and variability before interpreting significance.
- Check whether the chosen test matches the study design and variable types.
- Use confidence intervals, effect sizes, residuals, and post-hoc comparisons to explain the overall result.
- Treat non-significant results as insufficient evidence of a difference, not proof that groups are identical.
- Separate statistical evidence from broader causal or policy claims, particularly where samples are small.

Key reporting principle

A statistically significant result identifies evidence of a difference or association. The direction, size, practical meaning, and limitations must still be interpreted from the descriptive results and study design.

1. Anxiety Intervention Study

Independent-samples t-test comparing exercise and meditation

Research question and design

Twenty air traffic controllers were allocated to either regular exercise or meditation. Anxiety was measured after the intervention on a 0-100 scale. The analysis tested whether the two interventions produced different mean anxiety scores.

Null hypothesis: The mean anxiety score is equal in the exercise and meditation groups.

Alternative hypothesis: The mean anxiety score differs between the exercise and meditation groups.

Descriptive results

Group	n	Mean	SD	Variance
Exercise	10	65.4	3.17	10.04
Meditation	10	68.3	1.83	3.34

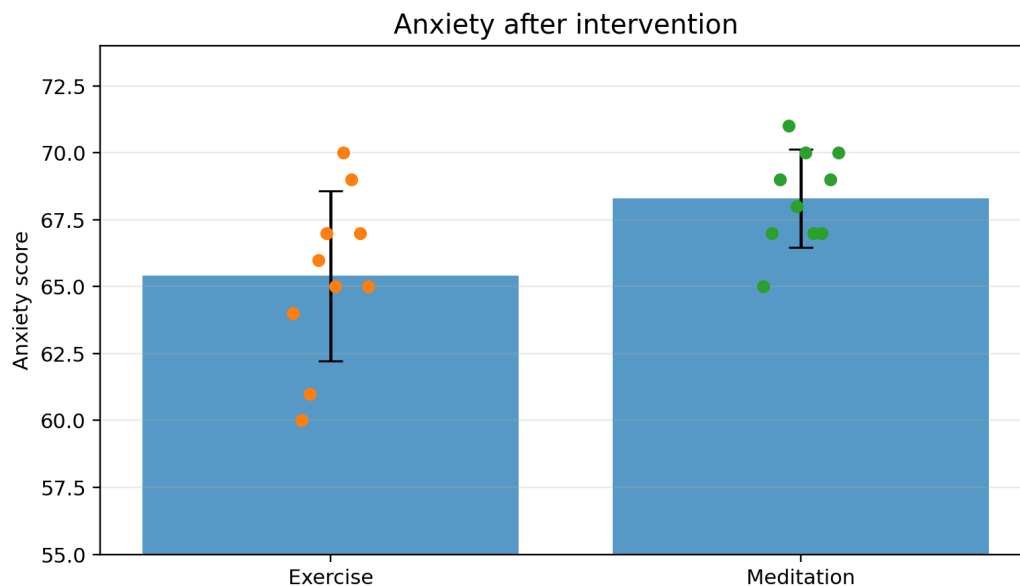


Figure 1. Group means with one standard-deviation error bars and individual observations.

Statistical result

The larger-to-smaller variance ratio was 3.00, below the conventional threshold of 4 used in the R analysis, so the pooled-variance t-test was retained. Exercise produced a mean anxiety score 2.9 points lower than meditation. The difference was statistically significant, $t(18) = -2.51$, $p = 0.022$, 95% CI for the exercise-minus-meditation difference $[-5.33, -0.47]$. Cohen's $d = -1.12$, indicating a large standardised difference.

Visual interpretation and conclusion

The chart shows substantial overlap between the two groups, but the exercise observations are shifted downward overall. The meditation scores cluster more tightly, while exercise scores are more dispersed. Because lower scores indicate less anxiety, the direction of the effect favours exercise.

Decision

Reject the null hypothesis. In this pilot sample, exercise was more effective than meditation at reducing post-intervention anxiety. The airline should prioritise exercise for a larger confirmatory trial rather than treating this small study as final policy evidence.

2. Personality Type and Film Preference

Chi-square test of independence with adjusted residual analysis

Research question and design

A sample of 120 participants classified as introverts or extroverts reported a preference for action films, dramas, or romantic films. The analysis tested whether film preference was independent of personality type.

Observed frequencies

Personality type	Action	Drama	Romantic	Total
Extrovert	10	25	30	65
Introvert	25	15	15	55
Total	35	40	45	120

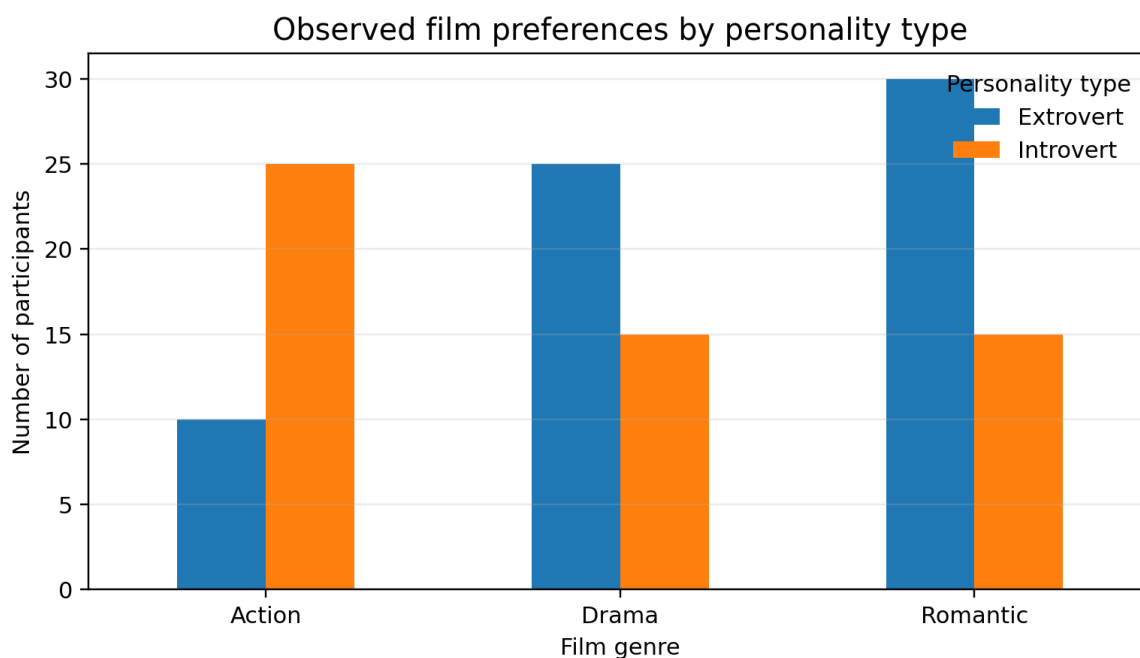


Figure 2. Observed counts show different preference profiles for introverts and extroverts.

Overall chi-square result

The association was statistically significant, $\chi^2(2) = 13.19$, $p = 0.0014$. Cramer's $V = 0.33$, indicating a moderate association. The null hypothesis of independence is rejected.

Expected frequencies under independence

Personality type	Action	Drama	Romantic
Extrovert	18.96	21.67	24.38
Introvert	16.04	18.33	20.62

2.1. Residual interpretation

Identifying which cells drive the overall association

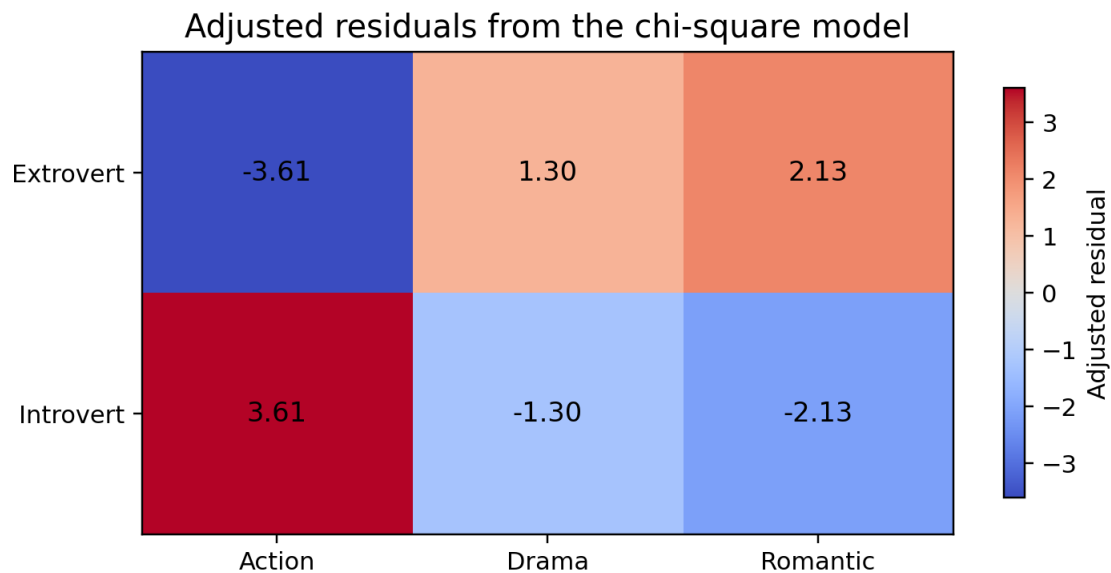


Figure 3. Adjusted residuals; absolute values above 1.96 indicate cells that differ significantly from expectation.

Adjusted residual findings

Cell	Adjusted residual	Interpretation
Extrovert - Action	-3.61	Significantly fewer than expected
Introvert - Action	3.61	Significantly more than expected
Extrovert - Drama	1.30	No significant deviation
Introvert - Drama	-1.30	No significant deviation
Extrovert - Romantic	2.13	Significantly more than expected
Introvert - Romantic	-2.13	Significantly fewer than expected

Interpretation

The overall association is mainly driven by action and romantic preferences. Introverts selected action films more often than expected, whereas extroverts selected them less often. The reverse pattern appears for romantic films: extroverts selected them more often than expected and introverts less often. Drama preferences were close to the frequencies expected under independence.

Conclusion

Personality type and film preference were associated in this sample. The result describes a group-level pattern and should not be interpreted as a deterministic rule for predicting an individual participant's preference.

Limitations

The analysis uses broad personality and film categories and does not control for age, gender, cultural background, or frequency of film consumption. The association may also vary in a different sample.

3. Weight-Loss Programme Comparison

One-way ANOVA across four independent groups

Research question and design

Forty participants were randomly assigned to a control group or one of three diet programmes: low fat, low carbohydrate, or low calorie. Weight loss after eight weeks was compared across the four groups. Positive values indicate weight loss and negative values indicate weight gain.

Descriptive results

Programme	n	Mean kg	SD	Min	Max
Control	10	1.1	1.52	-1	3
Low fat	10	4.7	1.95	2	8
Low carbohydrate	10	4.2	2.30	1	8
Low calorie	10	6.8	1.32	5	9

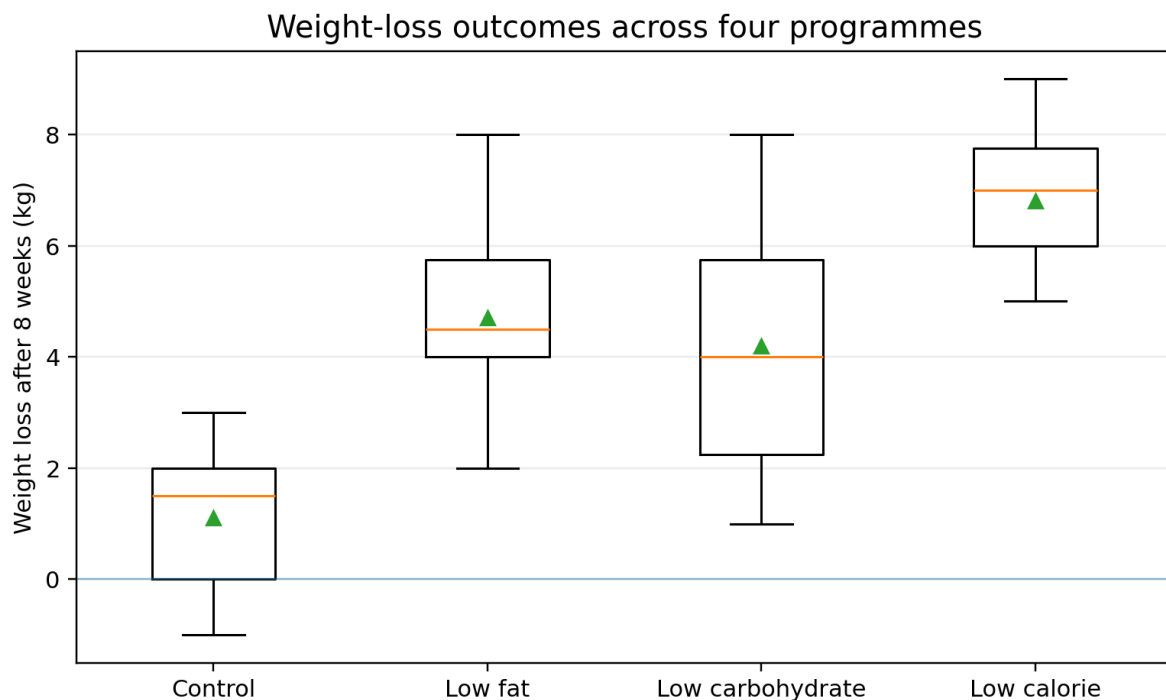


Figure 4. Distribution of weight loss by programme; triangles mark group means.

ANOVA result and effect size

Diet programme had a statistically significant effect on weight loss, $F(3, 36) = 16.87$, $p < .001$. Eta-squared = 0.58, meaning approximately 58.4% of the observed variance in weight loss was associated with programme assignment. This is a large effect.

Assumption checks

Assumption	Test result	Decision
Normality of residuals	Shapiro-Wilk $W = 0.982$, $p = 0.751$	No evidence of non-normality
Homogeneity of variance	Levene $F = 1.04$, $p = 0.386$	No evidence of unequal variances
Independence	Participants were randomly assigned	Supported by study design

3.1. Post-hoc comparisons

Determining which programmes differed after the significant ANOVA

Tukey HSD summary

Comparison	Mean difference (kg)	Adjusted p	Interpretation
Low calorie vs control	+5.7	< .001	Low calorie produced more weight loss
Low carbohydrate vs control	+3.1	.003	Low carbohydrate produced more weight loss
Low fat vs control	+3.6	< .001	Low fat produced more weight loss
Low calorie vs low carbohydrate	+2.6	.014	Low calorie produced more weight loss
Low calorie vs low fat	+2.1	.063	Not statistically significant
Low carbohydrate vs low fat	-0.5	.926	Not statistically significant

Visual interpretation and conclusion

The control group is centred close to 1 kg, with one participant gaining weight. All three diet programmes are shifted toward greater weight loss. The low-calorie programme has the highest mean at 6.8 kg and a relatively compact distribution. Low fat and low carbohydrate show similar central values and broader overlap.

Conclusion

Reject the null hypothesis. The programmes did not have the same effect. All three diets outperformed the control, and the low-calorie programme also outperformed the low-carbohydrate programme. The evidence did not distinguish low calorie from low fat at the .05 level.

Limitations

The analysis covers eight weeks and does not evaluate adherence, nutrition quality, adverse effects, or maintenance of weight loss. Statistical superiority in short-term weight loss does not by itself establish long-term safety or sustainability.

4. Fasting, Exercise, and Heart Health

4 x 2 factorial ANOVA with simple-effects comparisons

Research question and design

Forty participants completed one of four fasting conditions - control, 5/2, 48-hour fasting, or 16/8 - alongside frequent or infrequent exercise. The outcome was change in heart-health score over three months, with higher values representing greater improvement.

Cell means

Fasting programme	Frequent exercise	Infrequent exercise	Overall mean
Control	3.1	2.1	2.6
5/2	9.7	6.7	8.2
48-hour	2.9	6.5	4.7
16/8	8.3	6.3	7.3
Exercise mean	6.0	5.4	5.7

Factorial ANOVA

Effect	F	df	p	Partial eta-squared
Fasting programme	40.78	3, 32	< .001	0.79
Exercise frequency	2.27	1, 32	0.142	0.07
Fasting x exercise	13.40	3, 32	< .001	0.56

Fasting programme had a large significant main effect, exercise frequency did not have a significant overall main effect, and the significant interaction shows that the effect of a fasting programme changed according to exercise frequency. Because the interaction is significant, the simple effects are more informative than the overall exercise average.

Assumption checks

Residual normality was supported, Shapiro-Wilk $W = 0.968$, $p = 0.322$. Homogeneity across the eight cells was also supported, Levene $F = 0.56$, $p = 0.781$.

4.1. Interaction pattern

How exercise frequency changes the outcome of each fasting programme

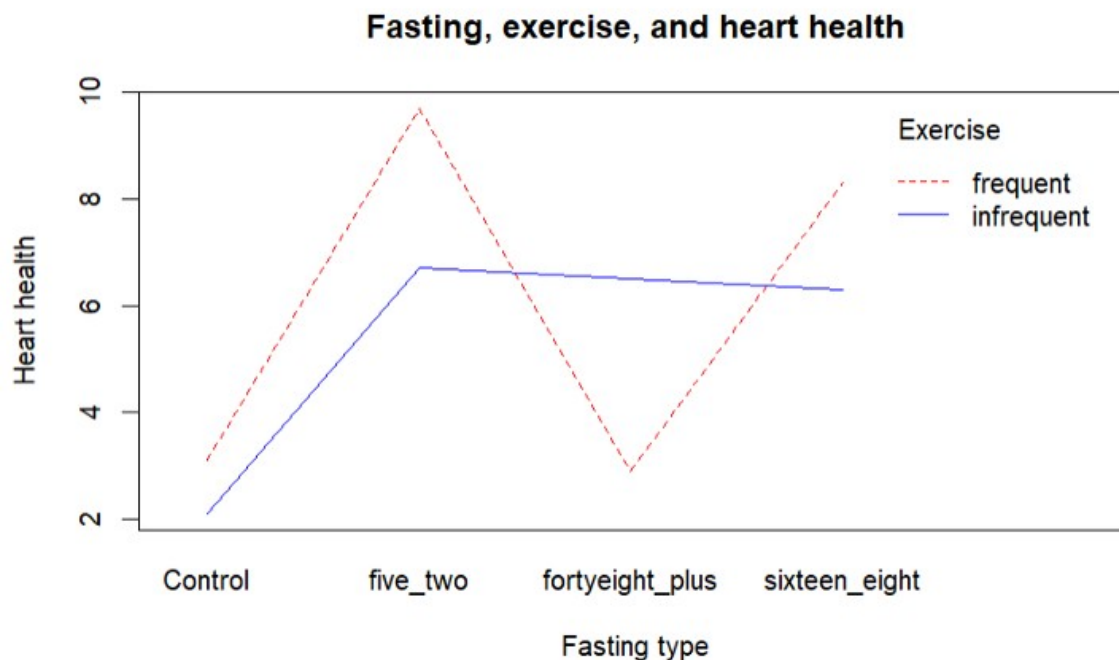


Figure 5. Interaction plot produced from the R analysis.

Visual interpretation

The lines are strongly non-parallel and cross between the 5/2 and 48-hour conditions, matching the significant interaction. Frequent exercise is associated with larger gains for 5/2 and 16/8, but not for 48-hour fasting. In the 48-hour condition, the pattern reverses: the frequent-exercise mean is 2.9 compared with 6.5 for infrequent exercise. The control means are low under both exercise conditions.

The highest cell mean is 9.7 for 5/2 with frequent exercise, followed by 8.3 for 16/8 with frequent exercise. The 48-hour frequent-exercise cell has a mean of 2.9, almost identical to the frequent-exercise control mean of 3.1. This indicates a lack of improvement relative to control rather than evidence that heart health became worse than baseline.

Interaction takeaway

Exercise cannot be interpreted as uniformly beneficial or harmful in this design. Its relationship with heart-health improvement depends on the fasting programme with which it is combined.

4.2. Simple-effects and risk interpretation

Tukey-adjusted comparisons within exercise and fasting conditions

```
exercise = frequent:
contrast      estimate    SE df t.ratio p.value
Control - five_two      -6.6 0.797 32  -8.282 <.0001
Control - fortyeight_plus    0.2 0.797 32   0.251 0.9943
Control - sixteen_eight     -5.2 0.797 32  -6.526 <.0001
five_two - fortyeight_plus    6.8 0.797 32   8.533 <.0001
five_two - sixteen_eight     1.4 0.797 32   1.757 0.3123
fortyeight_plus - sixteen_eight -5.4 0.797 32  -6.777 <.0001

exercise = infrequent:
contrast      estimate    SE df t.ratio p.value
Control - five_two      -4.6 0.797 32  -5.773 <.0001
Control - fortyeight_plus   -4.4 0.797 32  -5.522 <.0001
Control - sixteen_eight    -4.2 0.797 32  -5.271 0.0001
five_two - fortyeight_plus    0.2 0.797 32   0.251 0.9943
five_two - sixteen_eight     0.4 0.797 32   0.502 0.9580
fortyeight_plus - sixteen_eight 0.2 0.797 32   0.251 0.9943
```

Figure 6. Tukey-adjusted fasting-programme comparisons within each exercise level from the R output.

Comparisons within frequent exercise

- 5/2 exceeded control by 6.6 points ($p < .0001$), and 16/8 exceeded control by 5.2 points ($p < .0001$).
- 48-hour fasting did not differ from control (difference = -0.2 points in the 48-hour-minus-control direction; $p = .9943$).
- 5/2 and 16/8 did not differ significantly from each other (difference = 1.4, $p = .3123$).
- Both 5/2 and 16/8 substantially exceeded 48-hour fasting (differences = 6.8 and 5.4 points; both $p < .0001$).

Comparisons within infrequent exercise

All three fasting programmes exceeded control: 5/2 by 4.6 points, 48-hour fasting by 4.4 points, and 16/8 by 4.2 points (all adjusted $p \leq .0001$). The three fasting programmes did not significantly differ from one another under infrequent exercise.

Exercise comparisons within fasting programmes

Fasting programme	Frequent - infrequent	Adjusted p	Interpretation
Control	+1.0	.2186	No significant exercise difference
5/2	+3.0	.0007	Frequent exercise produced greater improvement
48-hour	-3.6	.0001	Frequent exercise produced less improvement
16/8	+2.0	.0173	Frequent exercise produced greater improvement

Conclusion

The results do not show that fasting generally damages heart health. They do identify a potentially unfavourable combination: 48-hour fasting with frequent exercise produced no advantage over the control condition and performed substantially worse than 5/2 or 16/8 with frequent exercise. This combination warrants caution and further study.

Limitations

The outcome is a change score measured after three months, and each factorial cell contains only five participants. The analysis does not establish clinical safety, mechanism, or long-term cardiovascular risk. Any recommendation should be confirmed in a larger clinically supervised study.

Overall conclusions

What the four analyses demonstrate

Study	Primary finding	Practical interpretation
Anxiety intervention	Exercise mean was 2.9 points lower; $p = .022$	Exercise was the stronger pilot intervention, subject to replication
Film preference	$\chi^2(2) = 13.19$; $p = .0014$; $V = .33$	Action and romantic preferences varied by personality type
Weight-loss programmes	$F(3,36) = 16.87$; $p < .001$; $\eta^2 = .58$	All diets beat control; low calorie had the highest mean
Heart-health study	Significant fasting x exercise interaction; $p < .001$	The effect of exercise depended on the fasting programme

Analytical methods applied

- Data import and preparation from Excel files in R.
- Group-level descriptive statistics and distribution visualisation.
- Independent-samples t-testing with variance assessment, confidence intervals, and standardised effect size.
- Chi-square testing with expected frequencies, Cramer's V, and adjusted residual interpretation.
- One-way and factorial ANOVA with assumption checks, η^2 effect sizes, and Tukey-adjusted comparisons.
- Interpretation of interaction plots, cell means, and simple effects without overgeneralising beyond the design.

Final note

The analyses illustrate why statistical conclusions should combine numerical results, visual patterns, effect sizes, post-hoc evidence, and design limitations. A p-value alone is not a complete interpretation.